Euexia Writeup

Problem Statement

A major challenge in digital health is not app installation, but sustained user engagement. Real-world usage studies show that retention drops quickly even for mental health and behavior-change apps.

Two findings motivate Euexia. First, low long-term engagement in real-world mental health apps: Baumel et al. found that sustained use is limited, with median retention around **3.9% at day 15 and 3.3% at day 30 across 93 apps** [1]. Second, steep early abandonment in health behavior apps: Kidman et al. reported a strong early drop-off pattern, with a median of **about 70% of users abandoning within the first 100 days across 18 studies involving 525,824 participants** [2].

Together, these findings highlight a clear gap: many health apps provide information but fail to sustain behavioral follow-through. In post-consultation contexts, this can lead to missed medication adherence, delayed follow-ups, and weak continuity of care.

Euexia addresses this gap by combining agentic extraction of consultation instructions into structured daily tasks and gamified mechanics designed to sustain engagement beyond initial app novelty.

Approach

Gamified Experience

Euexia's front-end experience is designed around recurring motivation loops rather than one-time content consumption.

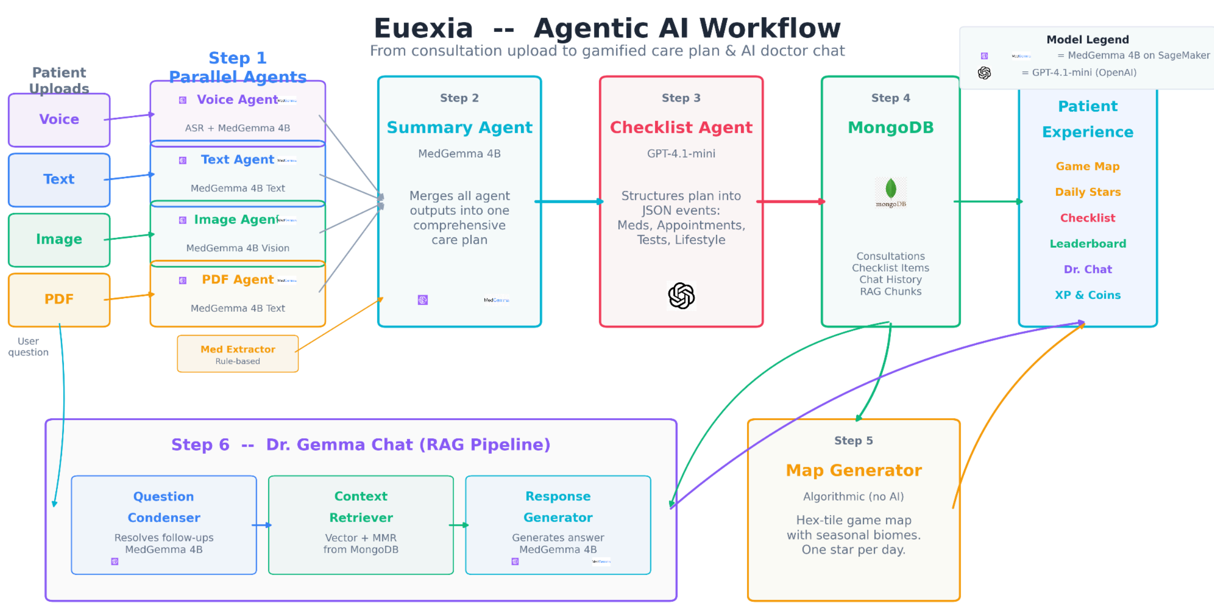
The map progression screen groups care tasks into day-based checkpoints, and completing tasks advances the player along the map, reinforcing visible progress by converting abstract adherence into daily wins.

The consultation upload screen allows users to upload voice, text, image, and PDF consultation content, lowering friction and supporting realistic patient behavior where medical information exists in multiple formats.

The store and leaderboard allow users to earn coins through completed tasks, unlock character skins, and compare progress socially to encourage long-term return behavior.

This design follows Yu-kai Chou's Octalysis framework [3], emphasizing accomplishment through checkpoint completion and map advancement, ownership through customizable characters and skins, scarcity through tiered pricing that creates delayed goals, and social influence through leaderboard comparison. The game layer is therefore not cosmetic but operationalizes adherence motivation through repeatable behavioral drivers.

Post-Consultation Agentic Workflow



The backend pipeline operates as an agentic sequence with explicit responsibility boundaries.

Input processing occurs in parallel where voice inputs undergo speech transcription and medical summarization, text inputs undergo summarization, image inputs undergo document interpretation, and PDFs undergo summarization. These summaries are merged into a single care-plan paragraph, which is then transformed into structured scheduled checklist events. Events are persisted in MongoDB and mapped into day-based checkpoints, and a RAG assistant ("Dr. Gemma") retrieves user-specific consultation and checklist context to generate follow-up responses.

Model responsibilities are clearly separated. MedGemma handles text summarization, image interpretation, and RAG chat generation. MedASR handles voice transcription. GPT-4.1 generates structured checklist events from the aggregated care-plan paragraph.

Deployment uses AWS SageMaker endpoints where the voice transcription endpoint runs on ml.c5.large (approximately 2 vCPU and 4 GiB RAM), while MedGemma endpoints run on ml.g5.2xlarge instances providing an NVIDIA A10G GPU (24 GB VRAM), 8 vCPU, and 32 GiB RAM suitable for multimodal medical workloads. This split keeps generation workloads GPU-backed while maintaining cost-efficient CPU processing for ASR.

Results

User Experience Speed

For a discharge summary upload example, observed backend timing was:

PIPELINE TIMING SUMMARY
0_db_create — 346 ms
1_uploads_parallel — 50,277 ms
2_aggregate — 50,553 ms
3_checklist_events — 91,860 ms
4_db_checklist — 93 ms
5_map_spec — 74 ms

TOTAL — 193,361 ms

The largest contributors are semantic processing stages including upload analysis, aggregation, and checklist generation, while database persistence and map translation overhead remain minimal.

For Dr. Gemma chat latency, prompts such as “how is my progress so far” and “Is there any food or liquids to avoid with my medicines?” both averaged approximately 16 seconds. These measurements indicate a functional but still optimizable experience for AI-heavy steps while orchestration overhead remains low.

References

[1] Baumel A, Muench F, Edan S, Kane JM. Objective User Engagement With Mental Health Apps: Systematic Search and Panel-Based Usage Analysis. Journal of Medical Internet Research. 2019;21(9):e14567. doi:10.2196/14567.

[2] Kidman PG, Curtis RG, Watson A, Maher CA. When and Why Adults Abandon Lifestyle Behavior and Mental Health Mobile Apps: Scoping Review. Journal of Medical Internet Research. 2024;26:e56897. doi:10.2196/56897.

[3] Chou Y-K. *Actionable Gamification: Beyond Points, Badges, and Leaderboards.* Octalysis Media; 2016.

Source code: https://github.com/elbochh/euexia